1 Introduction

Sub-Saharan Africa faces a significant waste management crisis. The consequences of improperly managed waste are severe, leading to environmental pollution, the spread of disease, and the degradation of community spaces. Formal collection systems are often overburdened, leaving communities to bear the brunt of the problem. However, this challenge also creates an opportunity for innovation.

Ayang, named from the Ibibio word for a sweeping broom, is a web-based platform that transforms waste collection into a rewarding, community-building activity. It uses gamification to motivate user participation , a method proven to drive pro-environmental behavior.

The technical core of Ayang is Google's Gemini Vision model. Users photograph a littered site before and after cleaning. The AI assesses the trash and verifies the cleanup, ensuring the integrity of the system. Points are awarded based on the estimated weight of the collected trash, directly incentivizing impactful cleanups. This project aligns with the DLI 2025 Community Challenge by offering a practical, AI-driven solution to a real-world African problem, designed to improve local environments and strengthen community bonds.

This document provides a condensed overview of the Ayang project. For a comprehensive review of our methodology, data, and technical architecture, please access the full report.

Read the full detailed report: link
Try the Ayang web app: `https://ayang-ai.vercel.app/`

2
Methodology

Ayang is built on a scalable architecture designed for an engaging user experience, comprising an AI model, a web application, and backend infrastructure.

Software Architecture: The user interface is a dynamic web application built with Next.js, enabling server-side rendering for fast performance even on low-bandwidth connections. The interactive map for defining community areas uses the Mapbox API. The backend is also handled by Next.js, using Server Actions to manage user authentication, image processing, and calls to the Gemini API, which simplifies the architecture. Data is stored in Neon, a serverless Postgres database platform.

AI & Image Analysis Pipeline: The AI-driven process is central to Ayang.

- A user uploads a "before" photo of a littered area with its GPS data.
- The backend sends the image to the Gemini Vision API to identify waste types and estimate their total weight in kilograms.
- After cleaning, the user uploads an "after" photo. The model performs a comparative analysis to verify the cleanup.
- Upon successful verification, points are calculated based on weight and awarded to the user.

Model Fine-Tuning: While the base Gemini model is capable, its accuracy can be significantly improved for niche tasks like weight estimation through fine-tuning. Our plan involves creating a custom dataset of local trash images. We will collect images of common local waste items, annotating each with a bounding box, trash type, and actual weight. This dataset will be used to fine-tune the model via Google's Vertex AI, making it more adept at recognizing local waste and estimating its weight accurately.

Community Features: To foster engagement, users can create communities by defining a location and radius on a map. Each community has a leaderboard to drive friendly competition. Future iterations will allow community leaders to schedule cleanup events.

3

Results and Findings

This section presents the expected outcomes and performance metrics for Ayang, as sufficient data has not yet been generated.

AI Model Performance:

- **Verification Accuracy:** We expect the base Gemini model to achieve over 95% accuracy in verifying cleanups, as this is a change detection task that large vision models handle well. The target for the fine-tuned model is over 99%.
- **Weight Estimation Accuracy:** This is more challenging. We project the base model's margin of error at ±30–40%. After fine-tuning with our custom dataset, we aim to reduce this error to under 15%.
- **Composition ID Accuracy:** We project an improvement from 85% accuracy on broad categories with the base model to 95% on specific local items with the fine-tuned model.

User Engagement: The platform's success depends on community motivation. Key performance indicators (KPIs) include:

- **User Growth:** Tracking monthly active users (MAU), with a target of 20% month-over-month growth for the first six months.
- **Cleanup Frequency:** Our target is an average of two cleanups per active user per month.
- **Leaderboard Activity:** High engagement with leaderboards will validate our gamification strategy.

Environmental Impact:

- **Total Waste Collected:** The platform will maintain a running total of the estimated weight of all collected trash. This will be a primary headline metric.
- **Waste Composition Data:** The AI's analysis will generate valuable data on waste types in different areas, which can be shared with municipal authorities to inform targeted interventions like the placement of recycling bins.

4

Challenges and Limitations

We acknowledge several challenges on the path to implementation.

Technical Challenges:

- **Image Recognition Accuracy:** The model's accuracy is core to Ayang's integrity. Poor image quality could lead to errors. Our mitigation strategy is user education and aggressive model fine-tuning.
- **Malicious Use:** Users could attempt to cheat the system, for instance, by dumping their own trash to photograph. We plan countermeasures like GPS verification, time-stamping, and AI-powered anomaly detection to flag suspicious activity.
- **Scalability and Cost:** User growth will increase API and storage costs. We will optimize the application and seek grants or partnerships to cover operational expenses.

Data and Ethical Challenges:

- **Dataset Bias:** Publicly available trash datasets may not represent waste found in African communities. A model trained on Western data may be less accurate locally. This is why creating a custom, localized dataset is a core part of our methodology.
- **Data Privacy:** The app collects location data, and we are committed to ethical handling. All user data will be anonymized for analysis, and our privacy policy will be transparent.

Logistical and Community Challenges:

- **The "Last Mile" Problem:** The app assists with collection but not final disposal. If no bins are nearby, trash may just be moved to another pile. We will partner with local governments or waste companies to schedule pickups at designated collection points.
- **Digital Divide:** Not everyone has a smartphone or affordable data, which may limit the app's reach. To promote inclusivity, we plan to explore offline functionality and community-based models where one person can document a group's work.

5 Recommendations & Next Steps

To build on the current prototype, we have a clear roadmap for future development.

Immediate Next Steps (3–6 Months):

- **Develop Custom Dataset:** Launch a data collection campaign for at least 5,000 annotated images of local waste. This is our highest priority for improving model accuracy.
- **Fine-Tune Model:** Use the new dataset to fine-tune the Gemini Vision model.
- **Launch Pilot Program:** Roll out the Ayang web app to 3–5 pilot communities to gather user feedback and test assumptions.
- **Establish Partnership:** Initiate a formal partnership with a local municipal waste authority or recycling company to solve the ”last mile” disposal problem.

Medium-Term Goals (6–18 Months):

- **Full Public Launch:** Launch Ayang to the general public in a target city, accompanied by a marketing campaign.
- **Advanced Gamification:** Introduce features like team vs. team challenges, special achievement badges, and corporate-sponsored prizes to provide a potential revenue stream.
- **Data Dashboard for Authorities:** Develop a web-based dashboard for municipal partners with real-time data visualizations and analytics.

Long-Term Vision (2+ Years):

- **National and Regional Scaling:** Expand the platform to other cities and countries, adapting the AI model for each new locality.
- **Integration with Circular Economy:** Forge partnerships with manufacturers and recycling plants to create a closed-loop system, using Ayang’s data to predict the supply of recyclable materials.
- **Policy Advocacy:** Use the accumulated data to advocate for better waste management policies, such as extended producer responsibility (EPR) laws.